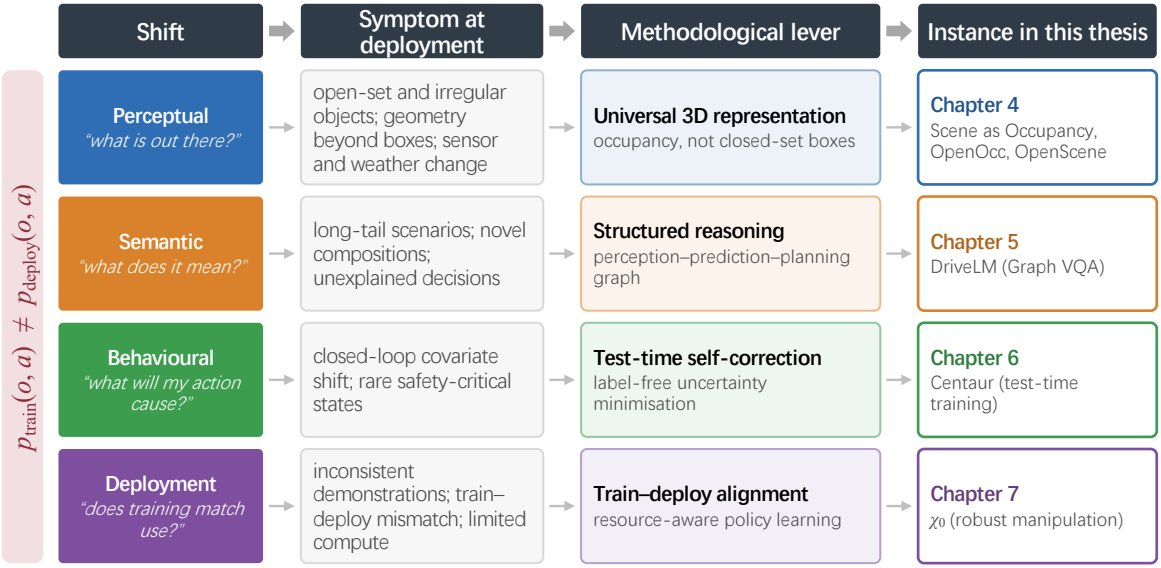

Shift
Symptom at deployment
Methodological lever
Instance in this thesis
$p_{\text{train}}(o, a) \neq p_{\text{deploy}}(o, a)$
Perceptual
"what is out there?"
open-set and irregular objects; geometry beyond boxes; sensor and weather change
Universal 3D representation
occupancy, not closed-set boxes
Chapter 4
Scene as Occupancy, OpenOcc, OpenScene
Semantic
"what does it mean?"
long-tail scenarios; novel compositions; unexplained decisions
Structured reasoning
perception–prediction–planning graph
Chapter 5
DriveLM (Graph VQA)
Behavioural
"what will my action cause?"
closed-loop covariate shift; rare safety-critical states
Test-time self-correction
label-free uncertainty minimisation
Chapter 6
Centaur (test-time training)
Deployment
"does training match use?"
inconsistent demonstrations; train–deploy mismatch; limited compute
Train–deploy alignment
resource-aware policy learning
Chapter 7
χ_0 (robust manipulation)